Exploring Chinese Healthcare System Outcomes among Different Groups

1. Introduction

As the world's second populous country, beating the third most populous by a wide margin, and an economy that has undergone rapid transformation in the past two decades, China has implemented a series of large-scale healthcare reforms since the early 2000s aimed at expanding insurance coverage, improving service delivery, and reducing financial barriers to care. With the government providing basic health insurance for at least 95% of the population, or 1.35 billion people, China covers almost an eighth of the world's population. These reforms provide insight into how a nation attempts to reconcile rapid economic development with the goal of ensuring equitable health outcomes.

Chinese health care is based on the public Basic Medical Insurance (BMI), which covers 1.35 billion people, or 95% of China's population, with add-on supplements such as commercial coverage, charitable help, and medical mutual aid (Yi, 2021). By focusing research on the BMI, scholars can discover how such a massive scale of coverage can be achieved while balancing the pros and cons of the system. This BMI is further split into three different categories. The mandatory Urban Employee Medical Insurance (UEMI) covers the employed urban workers. The voluntary Urban Resident Medical Insurance (URMI) covers unemployed urban residents, including the jobless, self-employed, children, and the elderly. The voluntary New Rural Cooperative Medical Scheme (NCMS) is the insurance that covers the rural residents of China. The two types of public insurance: employee basic medical insurance (EBMI) covers 337 million people, and the residents basic medical insurance (RBMI), which covers both rural and urban areas, cover 1.014 billion people (Yi 2021). However, this has led to noticeable gaps in the types of coverage and levels of coverage. Yi argues that total medical insurance financing, which is a part of total health spending, accounts for about 2.5% of China's GDP and that more investment is needed in the health care system, with a much greater focus on lower-income people (2021). In our research on the scholarship about the Chinese healthcare system, we have found studies focusing on the large-scale effects of reform on the system, and we have found studies that have focused on how certain groups of people are affected by gaps in healthcare. Our paper aims to explore the gap in how those large decisions and choices ripple out and affect different individuals and groups.

In recent years, China has focused on reforming its government to provide health care to widen coverage to the most people and has insured a massive number of people. However, in doing so, it has left holes in coverage and benefits, and has insufficient funding to equally cover everyone. These gaps tend to neglect and penalize the most rural and poor citizens in China.

2. Background

2.1 Healthcare system indicators

Health expenditure has long been a central focus of research on healthcare systems. Total health expenditures (THE) represent the amount spent on health care and related activities, such as administration of insurance, health research, and public health, and they include expenditures from both public and private funds (McGough et al., 2024). Scholars generally agree that health expenditures can result in improved healthcare access, reinforcing human capital, augmenting productivity, and ultimately boosting healthcare outcomes and economic performance (Li et al., 2024).

To understand the growth of health expenditure, we need to examine its relationship to gross domestic product (GDP). Because GDP is an overall economic indicator that reflects a country's total income and economic strength, the ratio of THE to GDP is widely used to assess the level of national input and investment in health. In addition, the continuous growth of GDP leads to the improvement of residents' income level, and the demand for residents' health services increases, which in turn leads to an increase in THE. Therefore, the development of the national economy is strongly associated with rising health expenditure (Jia et al., 2021). As an internationally recognized indicator, THE's continuous growth as a share of GDP, however, has become a common concern in most countries. Scholars even project that in the upcoming years, health expenditure will further threaten financial sustainability (Jia et al., 2021).

2.2 Understanding China's healthcare model and outcomes through comparison with the U.S.

China's healthcare indicators are compared with those of the United States, considering their similarities in the scale of the economy and demographic pressure in health care, and aging population. Most importantly, two countries represent different health system models, particularly in their financing structures and the government's role in healthcare provision. The US uses a private healthcare model that leaves much of the spending burden on citizens and insurance companies, with the exception of Medicare and Medicaid. China, on the other hand, focuses on health as a public good and uses a universal model where the government provides health care for most of its citizens. The comparison between them allows us to examine how different healthcare systems affect healthcare spending, health outcomes, and balance health, wealth, and equity. According to the International Monetary Fund, China, the world's second-largest economy with a GDP of $18.31 thousand billion in 2022, spent 7.02% on healthcare that year. The United States, the world's largest economy with a GDP of $26.05 thousand billion, devoted 16.63% of its GDP on healthcare expenditure, ranking first in the world. In China, economic growth, an aging population, and rising drug costs have contributed to the continuous growth of health expenditure. Notably, THE has grown so considerably since economic reform started in 1978 that its growth rate has exceeded that of GDP - the average annual growth rate of THE in 2009–2018 was 14.45%, compared with the average annual growth rate of GDP of 11.12% (Jia et al, 2021).

The comparison between China and the U.S.'s healthcare capacity and utilization shows mixed results. By 2020, China had 2.0 practicing physicians per 1,000 people on average, compared to 2.6 in the U.S. The gap in nursing resources is even larger: 2.7 nurses per 1,000 population in China versus 11.7 in the U.S (Fang, 2020). However, China has a higher inpatient utilization, with 173 hospital discharges per 1,000 people, compared to 125 in the U.S. In addition, China has more hospital beds: 4.3 per 1,000 population compared to the U.S.'s 2.8. In terms of health outcomes, China has experienced substantial improvements alongside rising health expenditure. The United States, in contrast, despite spending 3.2 trillion on healthcare, vastly more than China's, the expenditures do not consistently translate into better health outcomes (CMS.gov). By 2022, life expectancy at birth reached 75 years for males and 80.5 years for females, with an infant mortality rate of five per 1,000 live births. In the U.S., the life expectancy for both is lower: 73.7 years for males and 79.1 years for females (Commonwealth

Fund). The infant mortality rate is 6 per thousand live births. This rate may also come from a higher adolescent fertility rate: 13.6 births per 1,000 adolescent females in the U.S., compared to 5.5 in China (CDC). China also has a lower maternal mortality rate, with 15.7 per 100,000 live births versus 22.3 in the U.S (World Bank Group).

In addition to differences in health expenditure and service utilization, the two countries also diverge in how they train and supply their medical workforce. In China, the number of physicians is not regulated nationally. All medical schools are public, and the government tries to encourage more people to complete medical education. Annual tuition ranges from $1,408 to $2,816, depending on the region, and they are heavily subsidized by the government (Fang, 2020). To ensure the supply of healthcare services in rural areas, the governments waive tuition and lower entrance qualifications for some medical students. Medical students who attend these education programs have to commit to working in rural or remote areas for at least six years after graduation (Commonwealth Fund). In contrast, medical schools in the U.S. are not all public; around 40% of them are private, and the medical education is substantially more expensive, with median annual tuition reaching $39,153 at public medical schools and $62,529 at private schools in 2019. Around 73% U.S. medical students graduate with considerable debt, averaging around $ 200,000, although federal loan forgiveness and scholarship programs are included, particularly for trainees willing to serve in underserved regions (Commonwealth Fund). These stark differences in educational cost, regulation, and workforce incentives help explain the difference in healthcare expenditure, as well as their effort to reduce disparities in rural areas.

2.3 Major challenges of the current healthcare system

The source of healthcare financing has long been a key challenge for many countries, including China. According to the International Classification for Health Accounts (ICHA), total health expenditures can be divided into three categories: government health expenditure, social health expenditure, and out-of-pocket health expenditure. Government health expenditure is the government's funds for healthcare, including both central and local government financial allocations and subsidies. Social health expenditure reflects the total amount of funds raised from the non-government sectors, primarily through various public health insurance programmes. Out-of-pocket (OOP) health expenditure is the private financing from household income, which includes direct payments to healthcare services that are not fully reimbursed by insurance programs (Zang et al., 2017).

Among these categories, social health expenditure is the largest driver of THE in China, accounting for about 45 percent in 2022. Government health expenditure plays a relatively smaller role, contributing 28.17% of total health expenditure in the same year (Zang et al., 2020). Within social health expenditure, according to Li et al, the largest share comes from social security expenditures, which are public social health insurance funds financed through payroll contributions from employers, employees, etc. Private health insurance also plays a role in the social health expenditure category. Compared with U.S. private health insurance, which accounts for 34% of total health expenditure, in China, the same only contributes to 5.9% of the total healthcare expenditure, and the benefit package is often defined by the local government (Dai et al., 2025). However, private health expenditure has grown substantially in recent years due to the government's robust support for it and the public's growing awareness of healthcare (Dai et al., 2025).

While out-of-pocket expenditure increased the least compared to government and social healthcare expenditure, in contrast to developed economies, the proportion of out-of-pocket expenditures is relatively high, with 33.6% of the total healthcare spending, compared to 11.1% in the U.S. by 2022 (Zhang et al., 2017). The proportion of government health expenditures in China is excessively low. This high reliance on household out-of-pocket spending places a heavy financial burden on residents and poses affordability challenges within the healthcare system (Li et al., 2024). In 2009, due to escalating medical expenses and growing concerns about household-level healthcare-related financial burden, the Chinese government began to implement the new health system reform, mainly to aim to reduce the cost burden and alleviate the "difficulty and high cost of getting medical treatment" (Wang, 2024). Therefore, despite the achievement of near-universal insurance coverage, China faces an issue of a continuing high share of out-of-pocket expenditure and insufficient equity in health financing (Wang, 2024). This raises a crucial question: why does China's OOP expenditure remain so high, and what does this say about inequality and gaps in financial protection within its healthcare system?

3. China's Healthcare System Strengths And Gaps

China has achieved a 95% coverage rate in its coverage, but each of its types of public insurance varies in quality and coverage level. Much of this has to do with the evolution of the reform and the costs of providing health care to people. As China has reformed its healthcare system, it has continued to push more resources into its system. From 1990 to 2020, China's health insurance spending increased by 96.57-fold, and per capita spending in 2020 is 78.17 times greater than in 1990 (Wang, 2024). This proportionally massive increase in healthcare has resulted in 7.02% of China's GDP being spent on healthcare. However, this does put it below the US's overinflated 16.63% and the OECD average of around 10 percent. While China has made a fund of CNY ¥2.44 trillion and BMI is regarded as a public health good, there remain gaps in the coverage compared to other countries with "developed social security systems(Yi, 2021)." The fact that China has so many more people, yet has a lower than average GDP spent on health, does raise concerns about funding for health care. In a later paper, Wang argues that while China's government spending has increased every year, and its growth is higher than the global average, its equity in health financing in China has "not been sufficient" with significant "spatial and temporal differences in China's health financing structure (2024)." Meaning that while China has been increasing its healthcare every year, the rate at which it has been increasing has been unequal across different areas. The meaning that the urban area health care is increasing more and at a faster rate than in poorer rural areas.

3.1 Advantages of China's healthcare system

While the system aims to provide care to an unprecedented number of people with a smaller percentage of its GDP than many similar economies, this does provide China with some advantages. As the biggest purchaser of drugs and care in China, the government serves as a monopsony to lower drug prices. This monopsony on care allows China significant power over negotiating down costs and making healthcare cheaper to provide. China's commercial health premium income has skyrocketed by 30%, and the Country has been investing in other types of insurance, including commercial insurance, to increase coverage (Yi, 2021). This additional type of insurance serves as a way to boost the Chinese economy while providing stronger convergence to those who are willing to pay for it. This shows that while China is investing substantial resources in its health care system, it is seeing returns on those investments. Scholars believe that more investment in the system, with a particular focus on lower-income people, will allow the benefits to become more prominent while also increasing the equity of the system (Yi, 2021).

3.2 Gaps in the basic insurance plans

While China's plan provides coverage to a large number of people, there is still a gap in the healthcare system that fails to fully support people with fewer resources. This is evident in the 2017 article by Zhang et al., who argue that, based on surveys of middle-aged and older Chinese people, there is a gap in insurance coverage. Their regressions show that respondents with lower income per capita, less education, older age, divorced/widowed women, and rural registration are less likely to be insured (Zhang et al., 2017). Zhang et al. argue that the urban insurance plans UEMI and URMI have higher reimbursement rates than the rural plan NCMS. The median reimbursement rate of the UEMI for inpatient costs is 66%, while that of NCMS is only 25% (Zhang et al. 2017). So when the rural users of NCMS go to the hospital and use their insurance, they are much less likely to have their costs covered, and when they are, less costs are covered. This inequity results in a greater burden on rural and lower-socioeconomic-status people. Another striking finding is that with 36% of the total Chinese population living in rural areas (see Appendix 2), NCMS contributes the most to China's universal coverage (Zhang et al., 2017). The fact that such a large share of the population is enrolled in the least generous insurance scheme gives insights into why the near-universal coverage still leaves many people under-protected. While this data is based on surveys and only speaks to middle-aged and older people, meaning it is not as unbiased as it could be, its similar findings to the other papers point to an emerging trend in the information. Other scholars take this information and delve deeper into the causes of unequal risk.

3.3 Limitations in benefit depths and weak coverage

While studies earlier argue for gaps in reimbursement, the large amounts of covered people have led China to be more selective about what is covered at all. Beijing's 2018 outpatient ceiling was only CNY 3,000, compared with CNY 200,000 for inpatient care. The depth of coverage is rather shallow - for instance, many dental services, such as cleanings, are not covered by BMI. In addition, long-term care services are not part of China's health insurance at all. High-cost services such as home care, hospice care, and durable medical equipment like wheelchairs and hearing aids are paid entirely out-of-pocket. Preventive services are covered under a separate program and are free only for children and the elderly, while other residents must pay the full cost. People can use out-of-network services, but this always comes with much higher copayments (Fang, 2020). Additionally, the coverage excludes rural and urban migrants and their families (Yao & Barber, 2010). This shallowness of benefits is what allows China to boast a 95% coverage rate as one of the most populous nations in the world, while limiting the spending of its GDP. While this is less of an issue for people able to afford supplements through their insurance, it strikes a deeper blow to more at-risk populations.

4. Regional Inequities in China's Healthcare System

4.1 Consequences of poor rural coverage

The poor coverage of the NCMS can have disastrous effects on rural communities and other vulnerable communities. Li et al. explored the consequences using the data from the Fourth

National Health Service Survey to find the determinants of catastrophic health expenditures (2024). While this does not directly affect health outcomes, this statistic holds value as it shows who is least covered by insurance. It also holds value because people who are worried about being bankrupted by medical costs might adopt a necessity-only view of health care. This is not directly dangerous, but it may result in people being less active in preventative care and setting themselves up to be sicker in the future. Li et al. found that people are more likely to face catastrophic costs if they live in a poorer or rural region, have elderly or chronically ill members, have someone hospitalized, or are enrolled in the New Rural Cooperative Medical Scheme (NCMS) rather than urban insurance plans (2024). This is because, while the coverage is broad and can encompass many diseases, the benefits do not fully cover the cost of the illness (Li et al., 2024). For example, according to Hai Fang, public insurance in China reimburses only up to a fixed ceiling, and any spending above that cap must be paid entirely out-of-pocket. These ceilings are much lower for outpatient services than for inpatient care. This risk of catastrophic health expenditures being so prevalent among poor and rural communities weakens the arguments of the national healthcare system being a public good and requires an understanding of how this risk came about.

4.2 Unequal funding structures across regions

One of the key methods to understanding inequities in healthcare is to understand where the funding comes from and where it is put. In 2010, Yao and Barber wrote a briefing note on China's health care system for the World Health Report. While an older piece, this research holds immense value for its detailed focus on how healthcare is funded and used. Barber and Yao argued that while the government aimed to fund basic health services and public health services, the minimum level of care differs from region to region, and that, for NCMS, the rural insurance, a larger share of financing is provided by individuals and county-level governments through tax revenue (2010). This means that the state funds all of the insurance to a degree. However, for rural areas, this insurance is covered by a larger percentage of their taxes instead of funding other expenses. Rural populations are having a portion of their taxes being subverted into funding their government health care, while still being the most likely to be put into debt or be unable to afford medical care. There is a fundamental contradiction about health being a public good when unequal allocation of funds favors the more wealthy population, who are more likely to have access to better care and are able to afford supplemental insurance. This understanding that the costs of universal health care are not evenly distributed is one that later research discusses.

4.3 Unbalanced healthcare resource allocation across regions

How China's health care system uses those funds has evolved. China had a hospital-focused health system in which public hospitals were the primary providers of health care and received the lion's share of resources, whilst the primary health care sector was less developed (Cai et al. 2023). This resulted in most health care being received from hospitals. This is good for urban areas, as their large population density allows many more people to live near hospitals. It has a myriad of benefits that allow the health care system to work at maximum efficiency. This would be the opposite in rural areas, where people would live further apart, leading to longer drives for medical needs. Combined with the weaker insurance for rural healthcare, this weakens healthcare in rural areas.

However, China, since 2009, added deductible copayments for primary care services, free-of-charge essential public health services, and the introduction of family physicians, all of which increased primary care utilization(Cai et al. 2023). The primary health care reforms disproportionately benefited vulnerable people and poorer regions, and although the impact was considered modest, they did help reduce inequalities (Cai et al. 2023). This goes to show that while this will not achieve total equity in the system by itself, investing in making care more accessible in China goes a long way towards filling the gaps in the system.

This idea of how to reduce disparities is further explored in a study of financial compensation in the health care system from a supply-and-demand perspective. This looks at funding and decides if China's health care benefits more from financial compensation on the demand-side or supply-side and ultimately concludes that in more wealthy areas, there is a bigger boost from the supply-side, with the demand-side being more beneficial for areas with less economic growth (Guo et al. 2024). The demand-side means to invest in lowering costs and making healthcare more accessible to the people in the poorer and more rural areas. Supply-side focus for the more wealthy means lowering costs and incentivizing innovation in the richer areas to create higher quality care. By investing in supply-side growth in urban areas, China's medicine quality will continue to rise, and by making that quality more affordable for rural inhabitants, the effects should continue to compound on each other.

5. Class Inequities in China's Healthcare System

5.1 Differing quality of care

The fact that there is a difference in affordability for hospitals depending on location and class also equates to differing levels of care. The class difference is explored indirectly through a study on decreasing mortality rates in high-level hospitals. In analyzing inpatient outcomes in top tertiary general hospitals in China, a study found that, as time has gone on and coverage and technology have increased, inpatient mortality has significantly decreased (Ma et al., 2015). This study uses inpatient mortality as its primary metric of care, and by analyzing these factors, doctors and hospitals can become more efficient. This is worth noting because mortality has decreased and health care costs have increased in Top Tertiary General Hospitals. Health care has gotten better, but it has gotten more expensive for all parties involved. This is amazing for people who are able to afford care. However, as Ma et al. point out, these hospitals are concentrated in larger cities such as Beijing and Shanghai (2015). This indirectly speaks to a gap between rural and urban areas in terms of accessibility, and this gap also extends to people who cannot afford top hospitals. Analyzing this in combination with Li et al. 's 2024 survey, which puts rural and poor people as most likely to be bankrupted by health costs. While better care is beneficial for all parties involved, the high costs can lead to high levels of consequences for users. Poorer people are forced to choose between better care and potential costs against the risk of waiting for care and problems potentially getting worse. A universal health care system is traditionally designed to prevent precisely this problem by allowing everyone a basic level of care.

5.2 Approaches to combat inequities in healthcare

China has been working to fill these gaps. China has been investing in integrating material Medica or traditional medicine into its healthcare system (Dang et al. 2016). This is healthcare that uses more traditional Chinese medicine, as opposed to more recent medical technology. This is becoming a large part of the spending in healthcare, with the goal of lowering costs and making healthcare more affordable(Dang et al. 2016). This would make healthcare more accessible for rural and less wealthy populations, as traditional medicine would be less dependent on technology and could take advantage of more readily available resources. Additional solutions involve China investing more resources into its Healthcare system. This can take the form of funding directly allocated to reimbursements, which would strengthen programs like NCMS and allow poor and rural people to make better use of their healthcare without fear of catastrophic costs. A different solution involves further investment in the infrastructure of health. By investing more into creating more doctors and hospitals and incentivizing caregivers to provide more care in rural areas, the gap can be closed. This would also come with streamlining bureaucracy to make care more accessible and stronger. By intentional investments in its healthcare system, China can take advantage of its considerable resources to make a stronger system.

6. Conclusion

6.1 Summary of findings

For the past almost three decades, China has conducted a series of ambitious healthcare reforms, aiming to provide healthcare for all. In this paper, we found that while China is making efforts, its healthcare benefits are not equally distributed to everyone, leaving rural and less wealthy residents in a vulnerable position. China's near-universal insurance coverage is delivered through three major schemes—NCMS for rural residents, UEMI for urban employees, and URMI for unemployed urban residents—each targeting a specific population group. However, all three basic medical insurance schemes lack depth, which is the range of services covered by the insurance, leading to heavy out-of-pocket healthcare expenditure among all residents. In addition, the most widely used program - NCMS- offers the least generous benefits compared to the urban insurance programs. Beyond insurance design, healthcare access, utilization, and quality sharply vary between different regions and socioeconomic groups. This is due to several factors. First, the hospital-centric system, which makes much of the country's spending flow into large urban hospitals, leaving rural and primary-care with limited capacity, fewer skilled providers, slower improvements in service quality, and less efficiency. The issue of low-quality generics and unaffordable medications also disproportionately hurts groups with chronic conditions and low-income. As a result, healthcare in China is expensive, particularly penalizing rural areas and those with fewer financial resources.

6.2 Implications

China has seen a continuous expansion of total health expenditure, improvements in health status, increasing government health expenditure, and declining out-of-pocket expenditure. However, insufficiency of healthcare funding is still acute in China; the WHO report shows that China's total health expenditure as a share of GDP was ranked 90th out of all WHO member countries in descending order (Li et al.), and it is below the world average, 10%. Therefore, China still has potential for greater investment and fiscal input into the healthcare sector. Targeted investments are critical to reducing the imbalance between people with different health schemes, particularly between urban and rural schemes. Increasing the proportion of funding for rural regions, strengthening preventive care and chronic disease management there, and supporting grassroots primary care institutions are crucial to ensuring that vulnerable populations benefit equally from China's rapid growth in GDP and overall health expenditure.

6.3 Limitations and future research

Our study could be further improved by providing more perspectives on the government's initiatives to combat disparities in healthcare utilization and quality. For example, China stands out in terms of the substantial efforts to expand physician training and medical education, but our analysis does not examine how these initiatives translate into improved healthcare equity in recent years. Further research is needed to investigate how strengthening health technology disproportionately benefits those with better access, and how policies could be changed to raise accessibility. In addition, studying the role of primary care in reducing reliance on expensive hospital care, as well as conducting case studies and data analysis from other countries on how strong primary care narrows healthcare inequities, would also provide valuable insights for a more comprehensive policy recommendation.

Ultimately, health and healthcare are always growing, learning, and evolving industries, but if consistent and deliberate efforts are not made, then they will always favor the wealthy and powerful while leaving the poor and uninfluential to the side.

Bibliography

Barber, Sarah and Yao, Lan. “Health insurance systems in China: A briefing note” World Health Report, Background Paper 37, World Health Organization, 2010. Accessed November 11, 2025, https://cdn.who.int/media/docs/default-source/health-financing/technical-briefs-background-papers/whr-2010-background-paper-37.pdf?sfvrsn=e7bf652f_3&download=true.

Cai, Chang, Shangzhi Xiong, Christopher Millett, Jin Xu, Maoyi Tian, and Thomas Hone. “Health and Health System Impacts of China’s Comprehensive Primary Healthcare Reforms: A Systematic Review.” *Health Policy and Planning* 38, no. 9, July 28, 2023: 1064–78. https://doi.org/10.1093/heapol/czad058.

Commonwealth Fund. International Health Care System Profiles: United States. June t, 2020, Accessed November 23, 2025. https://www.commonwealthfund.org/international-health-policy-center/countries/united-states.

Dai Xin, Ma Xin, *The growing role of commercial health insurance in China*, (July 2025). Accessed November 23, 2025. https://www.swissre.com/institute/research/topics-and-risk-dialogues/china/commercial-health-insurance-china.html.

Dang, Haixia, Qiong Wang, Hainan Wang, Ming Yan, and Xinmin Liu. "The Integration of Chinese Material Medica into the Chinese Health Care Delivery System, an Update." Phytotherapy Research 30, no. 2 (2016): 292-297. Accessed November 13, 2025. https://rave.ohiolink.edu/ejournals/article/329598928.

“Databank.” DataBank. Accessed November 23, 2025. https://databank.worldbank.org/.

Fang, Hai. *China*. Peking University, The Commonwealth Fund. June 5, 2020, Accessed November 23, 2025. https://www.commonwealthfund.org/international-health-policy-center/countries/china .

Guo, Yi, Xuezhi Hong, Dongmei Li, Qiannan An, Wenwen Fan, Minghe Yang, and Luyang Xiao. “The Economic Effect of Financial Compensation in China’s Healthcare System: Comprehensive Insights Regarding Supply and Demand Factors.” Health economics review, March 16, 2024. https://pmc.ncbi.nlm.nih.gov/articles/PMC10943810/.

“Health Care Expenditures - Health, United States.” Centers for Disease Control and Prevention, (5 August 2024). Accessed November 23, 2025. https://www.cdc.gov/nchs/hus/topics/health-care-expenditures.htm.

International Monetary Fund. *International Monetary Fund, GD, Current Prices, United States and People’s Republic of China*. IMF. (2025). Accessed November 23, 2025. https://www.imf.org/external/datamapper/NGDPD@WEO/CHN/USA

Jia, Huanhuan, Hairui Jiang, Jianxing Yu, Jingru Zhang, Peng Cao, and Xihe Yu. "Total Health Expenditure and Its Driving Factors in China: A Gray Theory Analysis." *Healthcare* 9, no. 2 (February 14, 2021): 207. Accessed November 23, 2025. https://doi.org/10.3390/healthcare9020207.

Li, Hong-yan, and Rui-xue Zhang. "Analysis of the Structure and Trend Prediction of China's Total Health Expenditure." *Frontiers in Public Health* 12 (September 24, 2024). Accessed November 23, 2025. https://doi.org/10.3389/fpubh.2024.1425716.

Li, Ye, Qunhong Wu, Ling Xu, David Legge, Yanhua Hao, Lijun Gao, Ning Ning, and Gang Wan. "Factors Affecting Catastrophic Health Expenditure and Impoverishment from Medical Expenses in China: Policy Implications of Universal Health Insurance." *Bulletin of the World Health Organization* 90, no. 9, September 1, 2012 : 664–71. https://doi.org/10.2471/blt.12.102178.

McGough, Matthew, Matthew McGough, Emma Wager Twitter, Aubrey Winger, Nirmita Panchal, and Lynne Cotter. "How Has U.S. Spending on Healthcare Changed Over Time?" Peterson-KFF Health System Tracker, (December 20, 2024.) Accessed November 23, 2025. https://www.healthsystemtracker.org/chart-collection/u-s-spending-healthcare-changed-time/#Total%20national%20health%20expenditures,%20US%20$%20per%20capita,%201970-2023.

Ministry of Education "Press Releases." Ministry of Education of the People's Republic of China. (March, 2023) Accessed November 23, 2025. http://en.moe.gov.cn/news/press_releases/202304/t20230403_1054080.html#:~:text=Higher%20education%20further%20expanded%20its,of%20all%20students%20in%20China

Office of the Actuary. "National Health Expenditures 2022 Highlights." CMS.gov. (13 December, 2023) Accessed November 23, 2025. https://www.cms.gov/newsroom/fact-sheets/national-health-expenditures-2022-highlights

Wang, Na. "The Temporal and Spatial Interpretation of China's Health Financing: What Do Chinese Government Do' in New Healthcare Reform?" *Health Economics Review* 14, no. 1 (September 17, 2024). Accessed November 23, 2025. https://doi.org/10.1186/s13561-024-00551-1.

Wang, N. (2024). The temporal and spatial interpretation of China's health financing: What do Chinese' government 'do' in new healthcare reform? *Health Economics Review*, *14*(1). https://doi.org/10.1186/s13561-024-00551-1 https://doaj.org/article/3c3d9c510d6045b4b26e3b7807ba1da8

WHO Data. "China." World Health Organization. (2025) Accessed November 23, 2025. https://data.who.int/countries/156.

World Bank Group. *Adolescent fertility rate (births per 1,000 women ages 15-19)*. World Bank Gender Data Portal. (2025). Accessed November 23, 2025. .https://genderdata.worldbank.org/en/indicator/sp-ado-tfrt?year=2022

Xie-Min Ma, Xiao-Hong Chen, Ji-Shan Wang, Gary H. Lyman, Zhi Qu, Jiang-Chen Song, Chuan-Kin Zhou, and Lue Ping Zhao. "Evolving Healthcare Quality in Top Tertiary General Hospitals in China during the China Healthcare Reform (2010–2012) from the Perspective of Inpatient Mortality." 2015. *PLoS ONE* 10 (12): 1–16. Accessed November 11, 2025 https://doi.org/10.1371/journal.pone.0140568.

Yi, Baokang. "An Overview of the Chinese Healthcare System." PubMed Central, February 10, 2021. Accessed November 11, 2025 https://pmc.ncbi.nlm.nih.gov/articles/PMC7867737/.

Zhang, Chuanchuan, Xiaoyan Lei, John Straus, and Yaohui Zhao. "Health Insurance and Health Care among the Mid-Aged and Older Chinese: Evidence from the National Baseline Survey of CHARLS." Health Economics, vol. 26, no. 4, 2017, pp. 431-449. OhioLINK Electronic Journal Center, rave.ohiolink.edu/ejournals/article/346167796.

Zang, Shuang, Meizhen Zhao, Jing OuYang, and Xin Wang. "Interpretation of China's 2017 Health Expenditure: A Latent Profile Analysis of Panel Data." *BMJ Open* 10, no. 6 (June 2020). https://doi.org/10.1136/bmjopen-2019-035512.

Appendix

Table 1

<table>
<tr><th>Indicators</th><th colspan="2">China</th><th colspan="2">US</th></tr>
<tr><td colspan="5">Demographics (Year 2022)</td></tr>
<tr><td>Total Population</td><td colspan="2">1.41 billion</td><td colspan="2">0.334 billion</td></tr>
<tr><td>Age 65+ (% of total population)</td><td colspan="2">14%</td><td colspan="2">17%</td></tr>
<tr><td>Rural population (% of total population)</td><td colspan="2">36%</td><td colspan="2">17%</td></tr>
<tr><td>GDP</td><td colspan="2">$18.31 thousand billion</td><td colspan="2">$26.05 thousand billion</td></tr>
<tr><td>GDP per capita</td><td colspan="2">$12.97 thousand</td><td colspan="2">$77.94 thousand</td></tr>
<tr><td colspan="5">Healthcare-related Spending (Year 2022)</td></tr>
<tr><td>Share of GDP spent on health expenditure</td><td colspan="2">7.02%</td><td colspan="2">16.63%</td></tr>
<tr><td>Health care spending per capita</td><td colspan="2">$898.60</td><td colspan="2">$12,434.43</td></tr>
<tr><td>Out-of-pocket health care spending per capita</td><td colspan="2">$225.86</td><td colspan="2">$1380.12</td></tr>
<tr><td>Out-of-pocket expenditure
(% of current total health expenditure)</td><td colspan="2">26.9%</td><td colspan="2">11.1%</td></tr>
<tr><td>Government health expenditure
(% of current total health expenditure)</td><td colspan="2">28.17%</td><td colspan="2">45%</td></tr>
<tr><td>Health insurance coverage
(% of population)</td><td colspan="2">95%</td><td colspan="2">92%</td></tr>
<tr><td>Private health insurance
(% of current total health expenditure)</td><td colspan="2">5.9%</td><td colspan="2">34%</td></tr>
<tr><td>Annual tuition for medical school
(Year 2020)</td><td colspan="2">$1,408 - $2,816</td><td colspan="2">$39,153 - $62,529</td></tr>
<tr><td colspan="5">Health System Capacity & Utilization (Year 2020)</td></tr>
<tr><td>Practicing Physicians
per 1,000 population</td><td colspan="2">2.0</td><td colspan="2">2.6</td></tr>
<tr><td>Nurses per 1,000 population</td><td colspan="2">2.7</td><td colspan="2">11.7</td></tr>
<tr><td>Hospital beds per 1,000 population</td><td colspan="2">4.3</td><td colspan="2">2.8</td></tr>
<tr><td>Hospital discharges
per 1,000 population</td><td colspan="2">173</td><td colspan="2">125</td></tr>
<tr><td colspan="5">Health Status & Disease Burden (Year 2022)</td></tr>
<tr><td rowspan="2">Life expectancy at birth by sex</td><td>Female</td><td>Male</td><td>Female</td><td>Male</td></tr>
<tr><td>80.5</td><td>75</td><td>79.1</td><td>73.7</td></tr>
<tr><td>Adolescent fertility rate
(births per 1,000 women ages 15-19)</td><td colspan="2">5.5</td><td colspan="2">13.6</td></tr>
<tr><td>Infant mortality rate
(per 1,000 live births)</td><td colspan="2">5</td><td colspan="2">6</td></tr>
</table>

Indicators	**China**	**US**
Maternal mortality rate (per 100,000 live births)	15.7	22.3
Obesity Prevalence	6.6%	40.0%
Other Categories (2022)		
Birth rate (births per woman)	1	1.7
Enrollment rate for higher education	59.6%	62%
Adult Literacy rates (age 25-64)	Male: 99.3, Female: 98.1	Both genders: 99
Share of GDP spent on education (2020)	4.22%	5.22%